1.2 Analytic Techniques for Finding Limits of Functions

Name _____ Period ____ Date _____

AP Calculus 1.2 · Student sheet

I can…

- use the properties of limits and direct substitution.
- find a limit of the form 0/0 by factoring or by using a conjugate.
- find the limit of a difference quotient.

Vocabulary · fill in the missing word

Direct substitution — for a _____ P, the limit of P(x) as x → c is P(c); the same is true for a rational function if the denominator is not 0 at c

Indeterminate form 0/0 — the result of _____ that gives no information: more work is needed

Conjugate — _____

Part 1 · Same graph?

Graph $y = (x^2 - x - 6)/(x - 3)$ and $y = x + 2$.

	At x = 3	Near x = 3	Limit as x → 3
$(x^2 - x - 6)/(x - 3)$			
$x + 2$			

Part 2 · Notes and examples (fill in each note, then work the example)

Properties of limits: the limit of a sum, difference, product, or constant multiple is the sum, difference, product, or constant multiple of the _____.

Example 1

Find each limit.

a. as $x \to 3$: $2x^2 - x + 4$

b. as $x \to -2$: $(x - 1)(x + 4)$

Rational functions: the limit of R(x) as x → c is R(c) if the _____ is not 0 at c.

Example 2

Find each limit.

a. as $x \to 4$: $\sqrt{5x - 4}$

b. as $x \to 2$: $\dfrac{2x^2 + 5}{3x - 2}$

0/0: factor the numerator and the denominator, _____ the common factor, then substitute.

Example 3

Find the limit as $x \to 3$: $\dfrac{x^2 - 9}{x^2 - 3x}$

Radicals: multiply the numerator and the denominator by the _____.

Example 4

Find the limit as $x \to 4$: $\dfrac{\sqrt{x} - 2}{x - 4}$

Example 5

$f(x) = x^2 + 3x$. Find the limit as $h \to 0$ of $\dfrac{f(2 + h) - f(2)}{h}$.

Part 3 · Partner practice

Partner 1

Find the limit as $x \to -1$: $\dfrac{x^2 + 3x + 2}{x + 1}$

Partner 2

(AP® Practice) The limit as $x \to 5$ of $\dfrac{x^2 - 25}{x - 5}$ is

(A) 0 (B) 5 (C) 10 (D) nonexistent